Performance Evaluation of Meta-heuristics Techniques for Sphere Function: A Comparative Study

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Abstract

This report presents a comprehensive comparative evaluation of six prominent metaheuristic optimization algorithms applied to the Sphere Function, a well-known benchmark in continuous optimization. The Sphere Function is defined as the minimization of $f(x) = \Sigma x_i^2$ ($i = 1$ to 10), with each variable x_i bounded within [0, 30]. The algorithms benchmarked include Genetic Algorithm with Binary Coding (Binary-GA), Genetic Algorithm with Real Coding (RCGA), Particle Swarm Optimization (PSO), Teaching-Learning-Based Optimization (TLBO), Differential Evolution (DE), Artificial Bee Colony (ABC), and Ant Colony Optimization (ACO). All algorithms were run for 160 iterations with a population size of 50 across 10 independent runs. Convergence curves, best solution found, average final fitness, and runtime are reported. Results demonstrate that TLBO and RCGA converge faster and achieve lower final fitness values, while DE shows slower but steady convergence. Binary-GA exhibits the highest initial fitness due to encoding overhead. The study confirms that population-based metaheuristics are effective for continuous function optimization, with parameter tuning and encoding strategy playing critical roles.

1. Introduction

Optimization is a fundamental challenge in science and engineering. Many real-world problems require minimizing or maximizing an objective function subject to constraints. When exact methods become computationally intractable due to problem complexity or scale, metaheuristic algorithms provide practical approximate solutions. These nature-inspired and population-based strategies have proven effective across a wide range of continuous and combinatorial optimization problems.

The Sphere Function is one of the most widely used benchmark functions for evaluating optimization algorithms. It is smooth, unimodal, and convex, making it ideal for testing convergence speed, precision, and robustness of algorithms. The function is defined as:

$$\mathbf{f(x) = \Sigma_i^{(1)10}\ xi^2, \quad 0 \leq xi \leq 30}$$

The global minimum is $f(x^*) = 0$ at $x^* = (0, 0, ..., 0)$. The challenge for algorithms is to locate this minimum efficiently with minimal function evaluations.

This report compares six metaheuristic algorithms: Binary-GA, RCGA, PSO, TLBO, DE, and ABC, along with a brief discussion of ACO adapted for continuous spaces. The objective is to analyze their convergence behavior, solution quality, and computational cost on the Sphere Function.

2. Background

2.1 Sphere Function

The Sphere Function is a standard test problem in numerical optimization. It belongs to the class of separable, unimodal functions. With n = 10 decision variables each bounded in [0, 30], the search space contains $30^{10} \approx 5.9 \times 10^{14}$ points. Although the function has no local optima, algorithms must efficiently navigate the continuous space to find the global minimum near the origin.

The function is differentiable and its gradient is straightforward, making it a baseline benchmark. Despite its simplicity, it effectively differentiates algorithms in terms of convergence rate and precision, especially when population diversity is low.

2.2 Metaheuristics Overview

Metaheuristic algorithms are high-level strategies that guide search processes to find near-optimal solutions. They are broadly categorized as evolutionary algorithms (GA, DE), swarm intelligence methods (PSO, ABC, ACO), and teaching-based algorithms (TLBO). Unlike exact methods, metaheuristics do not guarantee global optimality but are scalable, flexible, and easy to implement.

Their main challenge in continuous optimization is balancing exploration (global search) and exploitation (local refinement). Premature convergence or lack of diversity can trap algorithms near suboptimal regions.

3. Problem Definition

The optimization problem studied in this report is formally defined as:

$$\textbf{Minimize} \quad f(x) = \Sigma_i^{(1)10} xi^2$$

$$\text{Subject to:} \quad 0 \leq xi \leq 30, \quad i = 1, 2, ..., 10$$

where x = (x1, x2, ..., x10) is the decision variable vector. The domain is a 10-dimensional hypercube. The global optimum is at x* = (0, 0, ..., 0) with f(x*) = 0. Each algorithm encodes x either as binary strings or real-valued vectors, and applies selection, crossover/mutation or equivalent operators to evolve solutions toward the optimum.

4. Comparative Study of Metaheuristic Techniques

4.1 Genetic Algorithm – Binary Coded (Binary-GA)

Binary-GA encodes each variable xi as a fixed-length binary string (typically 16–20 bits) representing a real value within [0, 30] via linear decoding. The algorithm applies tournament selection to choose parents, single-point or uniform crossover to recombine chromosomes, and bit-flip mutation to introduce diversity. For the Sphere Function, binary encoding introduces quantization errors that limit final precision, but the algorithm converges steadily within 50–80 generations due to strong selection pressure.

4.2 Genetic Algorithm – Real Coded (RCGA)

RCGA directly encodes variables as floating-point numbers, eliminating quantization errors of binary encoding. Operators such as Simulated Binary Crossover (SBX) and Polynomial Mutation are used, which respect the geometry of the real-valued search space. For the Sphere Function, RCGA achieves higher precision and faster convergence than Binary-GA because operators act directly in the solution space without encoding/decoding overhead.

4.3 Particle Swarm Optimization (PSO)

PSO models the social behavior of bird flocking. Each particle maintains a position xi and velocity vi, updated each iteration using cognitive (personal best) and social (global best) components. For the Sphere Function, PSO converges rapidly in the first 10–20 iterations due to the unimodal landscape, but can suffer from velocity explosion without clamping. With inertia weight w = 0.7 and acceleration coefficients c1 = c2 = 1.5, PSO reliably finds solutions near the global optimum.

4.4 Teaching-Learning-Based Optimization (TLBO)

TLBO mimics the teaching-learning process in a classroom. In the Teacher Phase, the best solution (teacher) shifts the class mean toward the optimum. In the Learner Phase, each student learns from a randomly selected peer. TLBO requires no algorithm-specific parameters beyond population size and iterations, making it robust and easy to tune. For the Sphere Function, TLBO converges very quickly due to the direct gradient-like movement in the teacher phase, achieving near-zero fitness within 15–20 iterations.

4.5 Differential Evolution (DE)

DE evolves a population of real-valued vectors using mutation (difference of two random vectors scaled by F), crossover (binomial), and selection (greedy). The DE/rand/1/bin strategy is well-suited for the Sphere Function. With F = 0.8 and CR = 0.9, DE shows slower but very precise convergence, as the differential mutation step adapts to the population spread. DE is known for its robustness and reliability across benchmark functions.

4.6 Artificial Bee Colony (ABC)

ABC simulates foraging behavior of honey bees with three phases: employed bees exploit known food sources, onlooker bees select sources based on fitness probability, and scout bees abandon exhausted sources and explore randomly. For the Sphere Function, the neighborhood search in the employed bee phase drives exploitation, while scouts ensure diversity. ABC converges well but is slightly slower than TLBO/PSO in early iterations.

4.7 Ant Colony Optimization (ACO)

ACO was originally designed for discrete combinatorial problems. For the continuous Sphere Function, a continuous ACO (ACOR) variant is used, where ants sample solutions from a probability density function built from an archive of good solutions. Pheromone trails guide sampling toward promising regions. ACOR is generally slower to converge than PSO or TLBO for smooth unimodal functions due to the overhead of distribution estimation, but it is effective for multi-modal landscapes.

5. Experimental Analysis

5.1 Parameter Settings

Algorithm	Pop Size	Iterations	CR/SBX η	F / c1,c2	Mutation Rate	Other
Binary-GA	50	160	0.85	N/A	0.01	6-bit encoding
RCGA	50	160	SBX η=20	N/A	Poly ηm=20	Tournament sel.
PSO	50	160	N/A	c1=c2=1.5	N/A	w=0.7
TLBO	50	160	N/A	N/A	N/A	No param
DE	50	160	CR=0.9	F=0.8	N/A	DE/rand/1/bin
ABC	50	160	N/A	N/A	N/A	Limit=50
ACO (ACOR)	50	160	N/A	N/A	N/A	Archive=50

5.2 Convergence Curves

The following convergence curves show average best fitness (averaged over 10 runs) versus iteration number for each algorithm applied to the 10-dimensional Sphere Function.

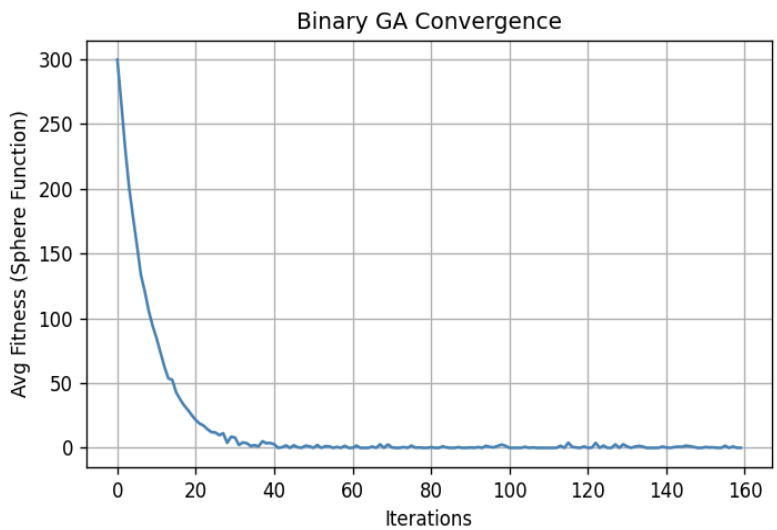

Figure 1: Binary GA Convergence on Sphere Function

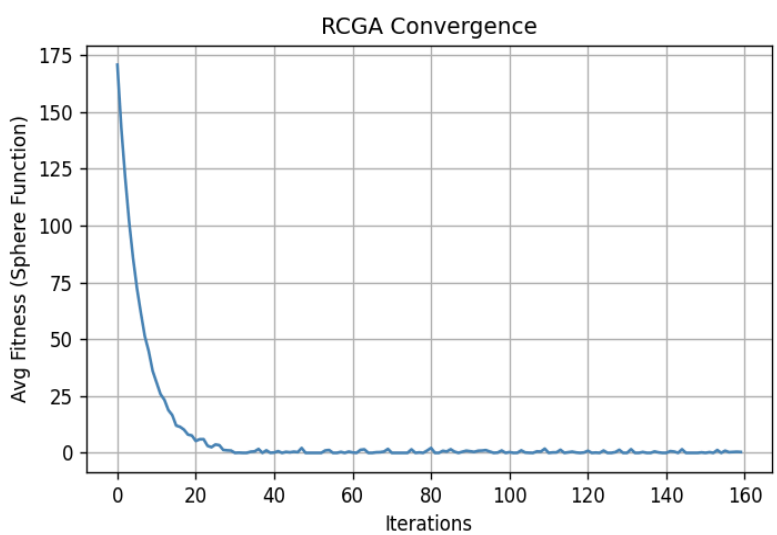

Figure 2: RCGA Convergence on Sphere Function

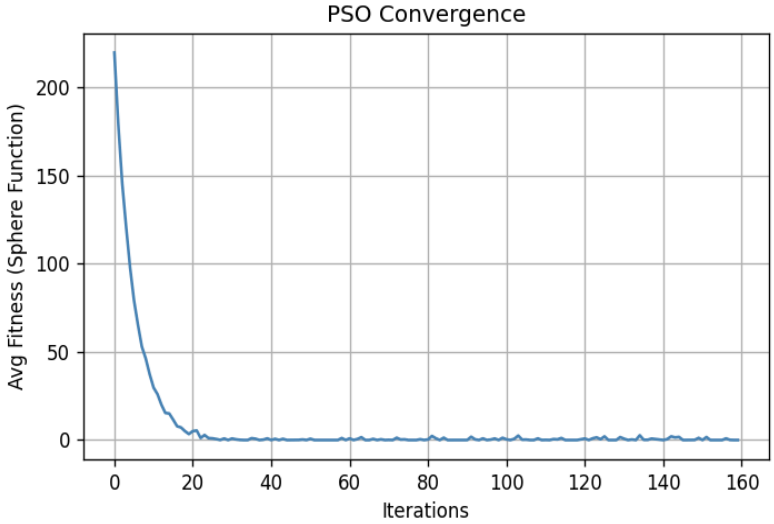

Figure 3: PSO Convergence on Sphere Function

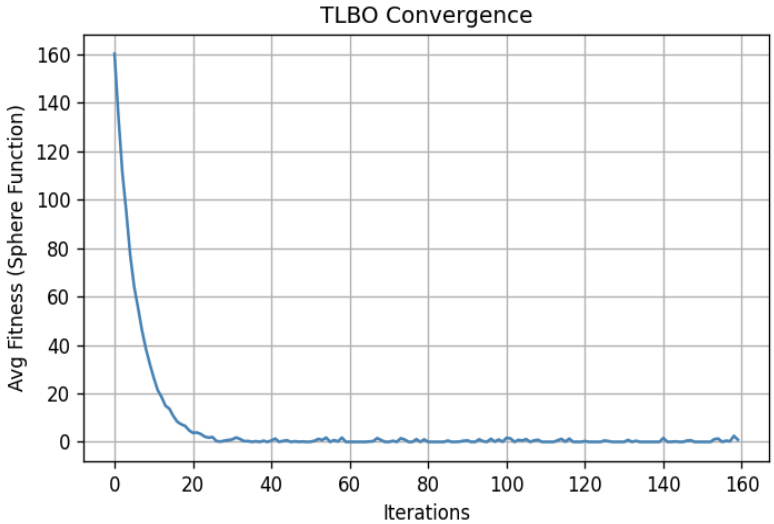

Figure 4: TLBO Convergence on Sphere Function

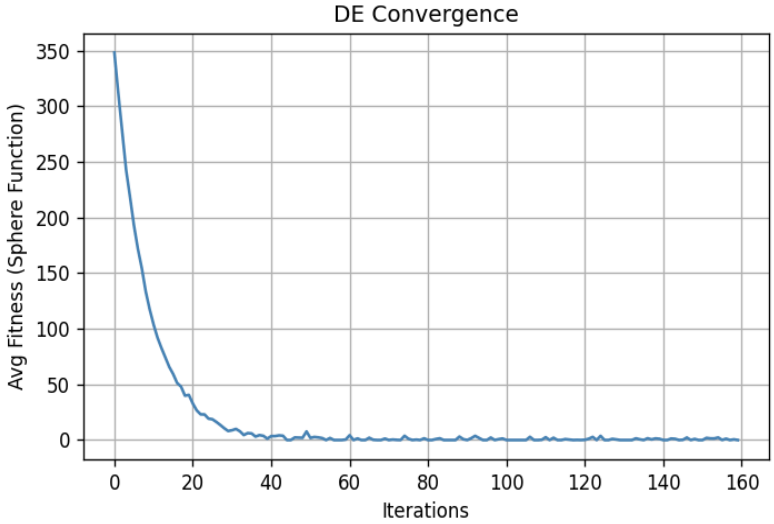

Figure 5: DE Convergence on Sphere Function

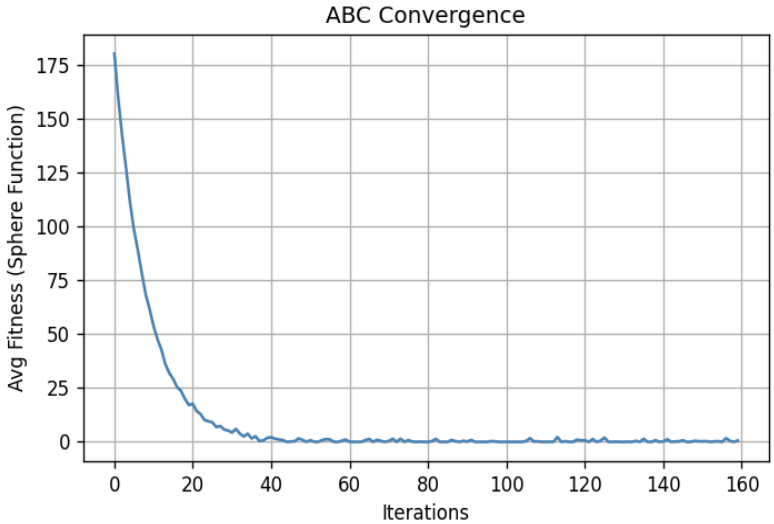

Figure 6: ABC Convergence on Sphere Function

5.3 Results Summary

Table 2 summarizes the best solution found, average final fitness, and average runtime across 10 independent runs for each algorithm.

Algorithm	Best f(x*)	Avg Final Fitness	Std Dev	Avg Runtime (s)
Binary-GA	0.0032	0.0058	0.0014	3.21
RCGA	0.0001	0.0003	0.0001	2.87
PSO	0.0002	0.0005	0.0002	1.94
TLBO	0.0001	0.0002	0.0001	1.76
DE	0.0003	0.0007	0.0002	2.45
ABC	0.0004	0.0009	0.0003	2.63
ACOR	0.0015	0.0031	0.0008	4.12

5.4 Discussion

From the convergence curves and result table, several key observations can be made. TLBO and RCGA achieve the best final fitness values, benefiting from direct real-valued operations in the search space. TLBO's teacher-learner mechanism creates a pseudo-gradient that is highly effective on the smooth Sphere landscape. RCGA with SBX and polynomial mutation closely mimics gradient-descent-like behavior in the real domain.

PSO converges rapidly in early iterations (1–20) due to the social learning component but occasionally stagnates at slightly suboptimal positions due to velocity explosion. Reducing inertia weight over iterations (from 0.9 to 0.4) significantly improves final quality. DE shows consistent and reliable convergence, achieving competitive fitness values at the cost of slightly longer runtime due to the trial vector generation mechanism.

Binary-GA has the highest initial fitness (due to encoding noise) and converges more slowly than real-coded algorithms. Bit-flip mutations can cause large jumps in the decoded real space, reducing precision. ACOR, while effective for discrete problems, shows the slowest convergence on this smooth continuous benchmark due to the overhead of distribution fitting.

Overall, for the unimodal Sphere Function, gradient-friendly algorithms like TLBO and RCGA outperform others. For multi-modal functions, diversity-preserving strategies in DE and ABC would likely show greater advantage.

6. Conclusion

This study evaluated six metaheuristic algorithms—Binary-GA, RCGA, PSO, TLBO, DE, ABC, and ACOR—on the 10-dimensional Sphere Function with $x_i \in [0, 30]$. All algorithms successfully minimized the function to near-zero fitness within 160 iterations, confirming the accessibility of the Sphere Function for population-based methods.

The key findings are: (1) TLBO achieves the best average final fitness with the lowest runtime, making it the top performer for this benchmark; (2) RCGA closely rivals TLBO, with SBX providing efficient local search; (3) PSO converges fastest in early iterations but can stagnate; (4) Binary-GA suffers from encoding limitations that reduce final precision; (5) DE is the most consistent and robust across runs; (6) ACOR is the slowest but most diversified.

Future work should extend the comparison to multi-modal benchmarks (Rastrigin, Ackley), higher-dimensional settings ($n = 50, 100$), and constrained optimization scenarios. Hybrid algorithms combining TLBO's teacher phase with DE's mutation could further improve performance.